

BIOGEOCHEMICAL CYCLES

Lesson 4

Objectives



Understand how **Biogeochemical Cycles** are interconnected and their importance to ecosystem health.



Analyze and explain diagrams or models of the cycles of matter, identifying key processes.



Develop an appreciation for the delicate balance of ecosystems and the role that the cycles of matter play in maintaining this balance.

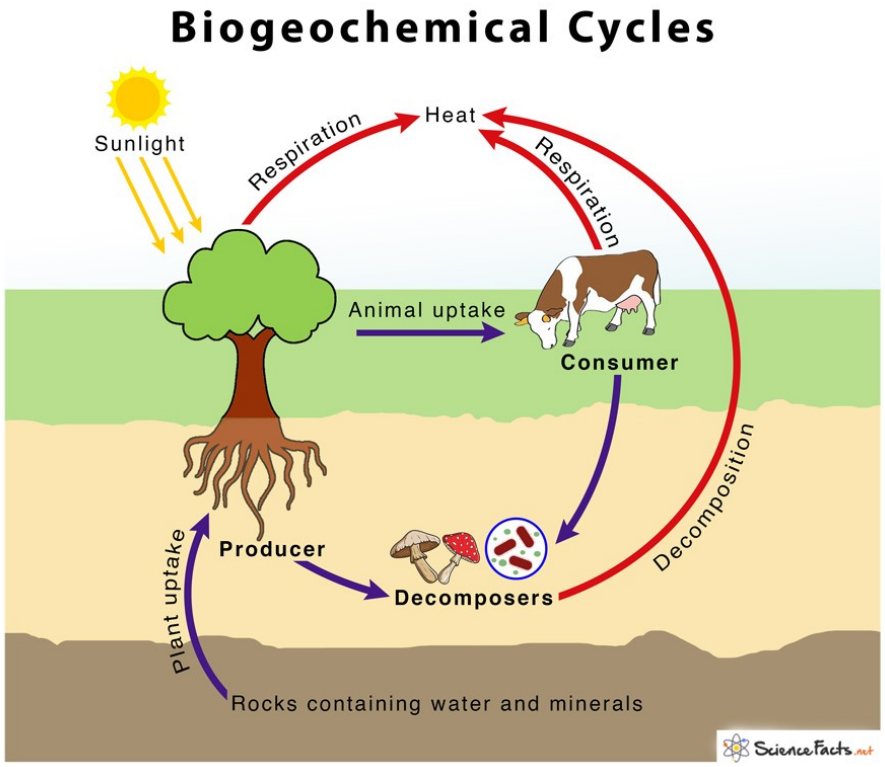
Biogeochemical Cycles

Biogeochemical Cycles are the **pathways** through which **chemical elements and compounds move** through the **biotic (living) and abiotic (non-living) components** of an ecosystem.

These cycles are **essential for life** as they **ensure the continuous recycling** of nutrients and other elements **necessary for organisms to survive and thrive.**

Major Biogeochemical Cycles

1) **Water Cycle (Hydrologic Cycle)**
2) **Carbon Cycle**
3) **Nitrogen cycle**
4) **Phosphorus Cycle**
5) **Sulfur Cycle**

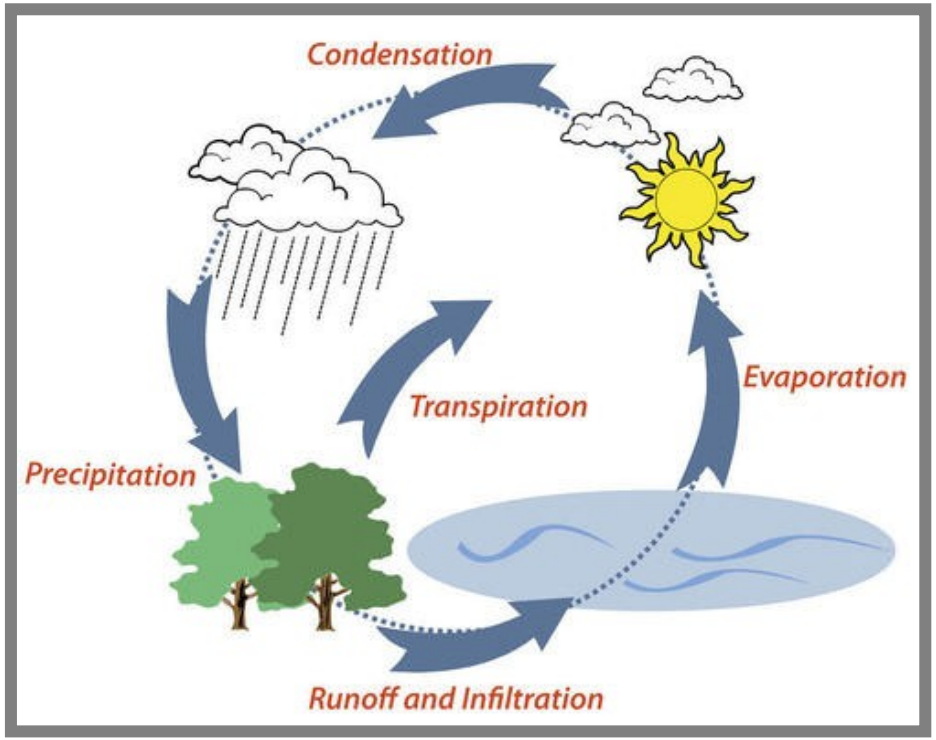


Water Cycle (Hydrologic Cycle)

The water cycle describes the continuous movement of **water** in its various forms (liquid, solid, gas) through the Earth's ***atmosphere, oceans***, and ***land.***

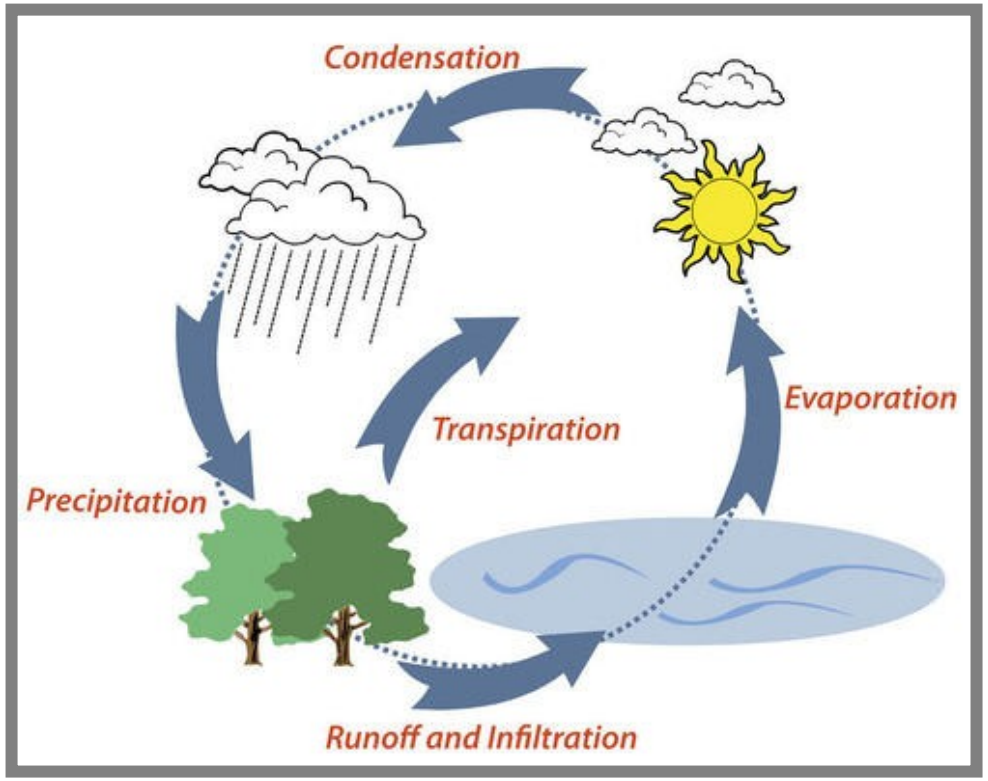
Essential for all life; influences climate, weather patterns, and the distribution of other elements.

Key Processes: Evaporation, transpiration, condensation, precipitation, infiltration, runoff.



Water Cycle (Hydrologic Cycle)

- **Evaporation**: The process by which water transforms from a liquid state to a vapor, rising into the atmosphere due to heat from the sun.
- **Transpiration:** The process of water movement through a plant and its evaporation from aerial parts, such as leaves, stems, and flowers. It is like the plant "sweating" out water vapor.
- **Condensation**: The transformation of water vapor back into liquid water droplets, **forming clouds** as the vapor cools and loses energy.
- **Precipitation**: The release of water from clouds in the form of **rain, snow, sleet, or hail** that falls back to the Earth's surface.
- **Infiltration**: The process by which water penetrates the soil surface and moves downward into the **groundwater supply**.
- **Runoff**: The flow of water, often from rain or melting snow, that travels across the land surface and eventually enters rivers, lakes, or oceans.

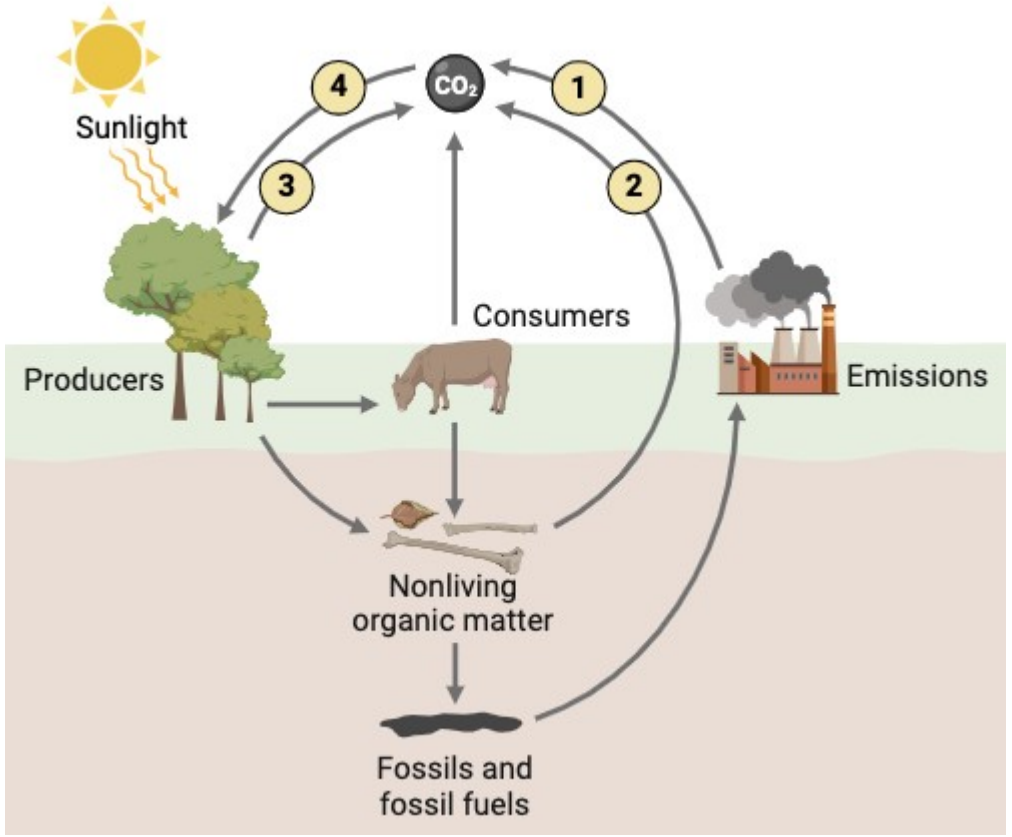


Carbon Cycle

The carbon cycle involves the movement of carbon through the atmosphere, oceans, land, and living organisms.

Carbon is the **backbone of all organic molecules**; **regulates Earth's temperature** and is a **key component of fossil fuels.**

Key Processes: Photosynthesis, respiration, decomposition, combustion, fossilization.

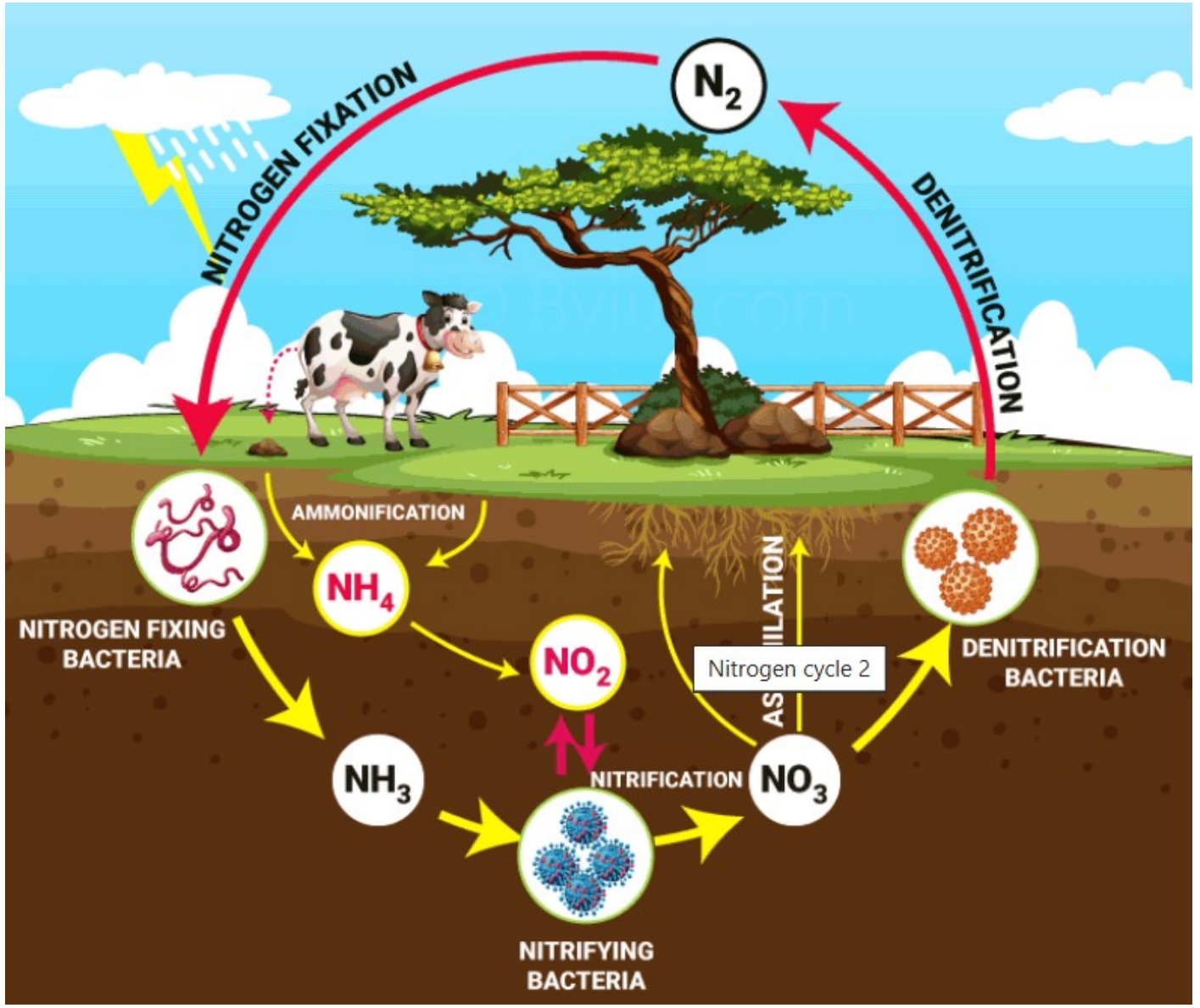


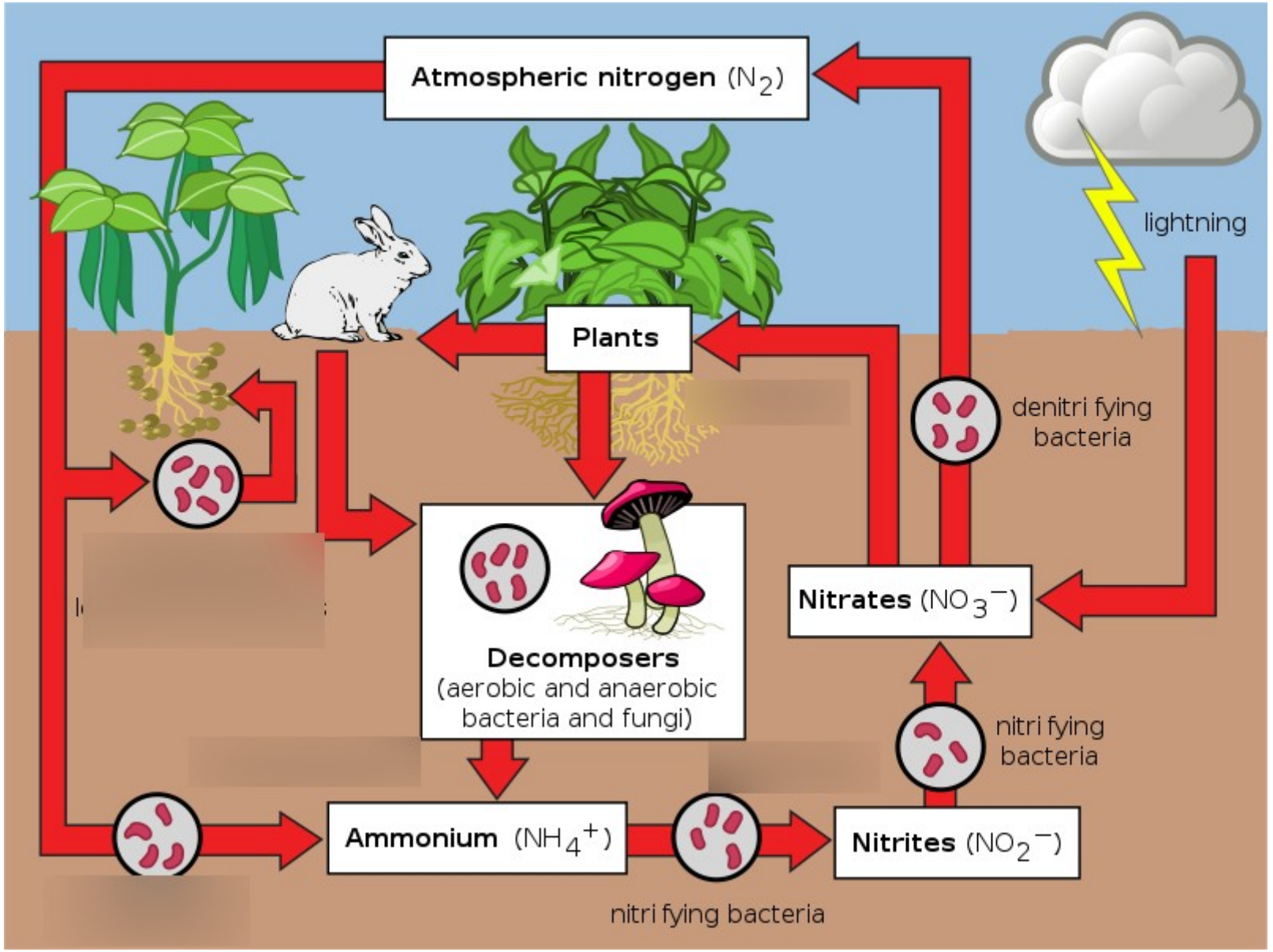
Carbon Cycle

- **Photosynthesis:** This is how plants (and some other organisms) use sunlight, water, and carbon dioxide in the atmosphere to create sugars (glucose) for energy and release oxygen as a byproduct.
- **Aerobic Respiration:** This is the process by which organisms (plants, animals, fungi, bacteria) break down sugars (like glucose) to release energy. This process consumes oxygen and releases carbon dioxide back into the atmosphere.
- **Decomposition:** When organisms die, their remains are broken down by decomposers (bacteria and fungi). This process releases carbon back into the soil and, as carbon dioxide, into the atmosphere.
- **Combustion:** Burning organic material (wood, fossil fuels) rapidly combines the carbon in the material with oxygen, producing carbon dioxide and releasing energy (heat and light).
- **Fossilization:** Over long periods of time, under specific conditions, the remains of dead organisms can be transformed into fossil fuels (coal, oil, natural gas). This process stores carbon in the Earth's crus

Nitrogen Cycle

- The nitrogen cycle is the process by which nitrogen is converted into various forms and cycled through the **atmosphere, soil**, and **living organisms**.
- Nitrogen is a crucial component of proteins and nucleic acids; often a limiting nutrient for plant growth.
- **Key Processes:** Nitrogen fixation, nitrification, denitrification, assimilation, ammonification.



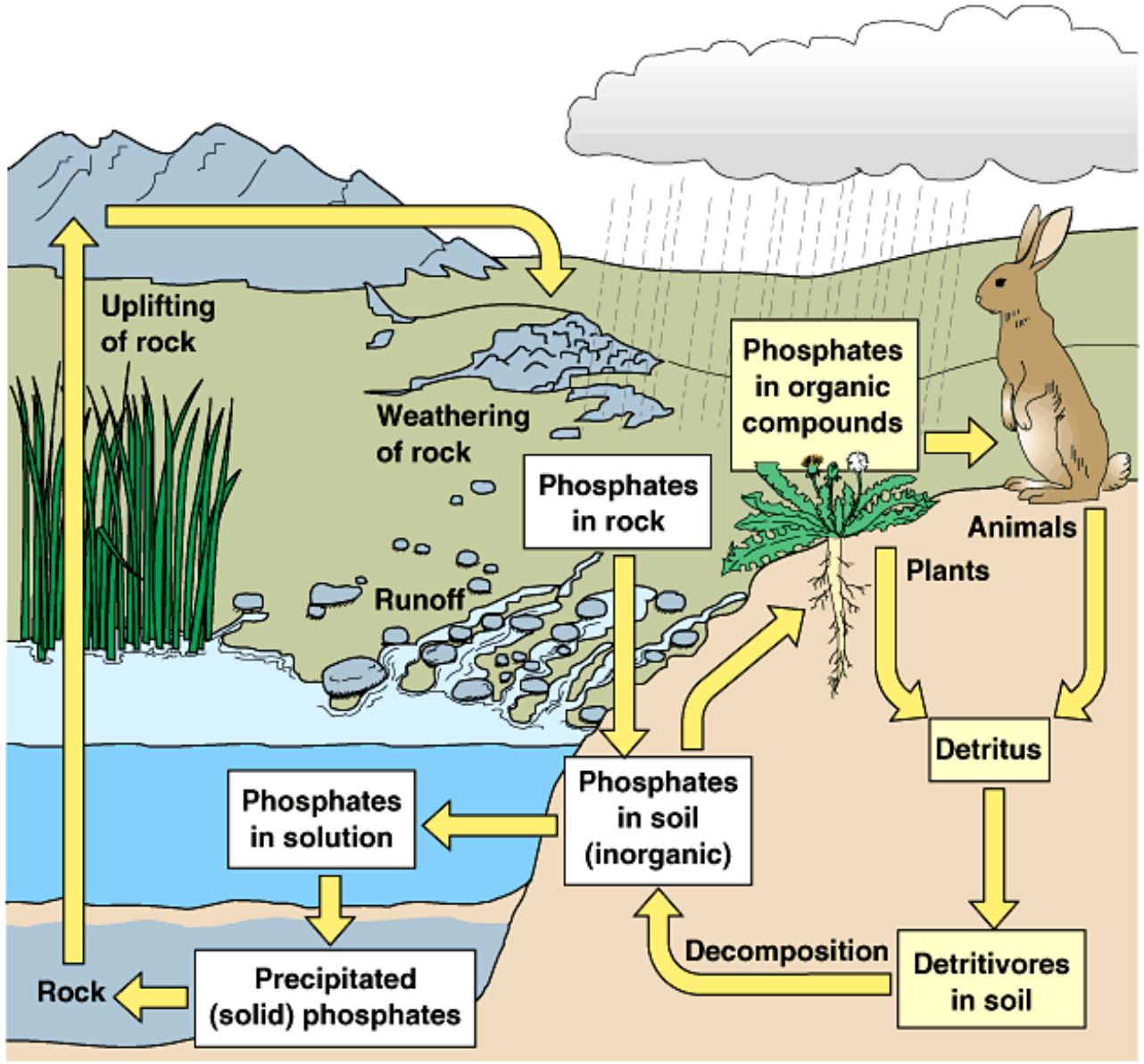

Atmospheric nitrogen (N_2)
lightning
Plants
denitri fying bacteria
Decomposers
(aerobic and anaerobic bacteria and fungi)
Nitrates (NO_3^-)
nitri fying bacteria
Ammonium (NH_4^+)
Nitrites (NO_2^-)
nitri fying bacteria

Nitrogen Cycle

- **Nitrogen Fixation:** This is the process where atmospheric nitrogen (N_2), which is unusable by most organisms, is converted into ammonia (NH_3) or other usable nitrogen compounds. This is primarily done by certain **bacteria** in the soil or in the roots of some plants (like legumes).
- **Nitrification:** Ammonia (NH_3) is converted into nitrites (NO^{2-}) and then **nitrates (NO^{3-})** by bacteria in the soil. **Nitrates** are a form of nitrogen that **plants can readily absorb**.
- **Denitrification:** Nitrates (NO^{3-}) are converted back into atmospheric nitrogen gas (N_2) by certain bacteria in oxygen-poor environments (like waterlogged soils). This process returns nitrogen to the atmosphere.
- **Assimilation:** Plants absorb nitrogen (in the form of nitrates or ammonia) through their roots and incorporate it into organic molecules like proteins and nucleic acids. Animals then assimilate nitrogen by eating plants (or other animals that have eaten plants).
- **Ammonification:** When organisms die or produce waste, the organic nitrogen in their remains is converted back into ammonia (NH_3) by decomposers (bacteria and fungi). This ammonia can then be used by plants or go through nitrification again.

Phosphorus Cycle

- The phosphorus cycle describes the movement of phosphorus through the Earth's ecosystems. Unlike other cycles, it **does not** have a significant **atmospheric component.**
- Phosphorus exists in nature as a phosphate ion (PO_4^{3-}).
- Phosphorus is essential for DNA, RNA, and ATP (the energy currency of cells); often a limiting nutrient for plant growth.
- **Key Processes:** Weathering, erosion, assimilation, decomposition, sedimentation.



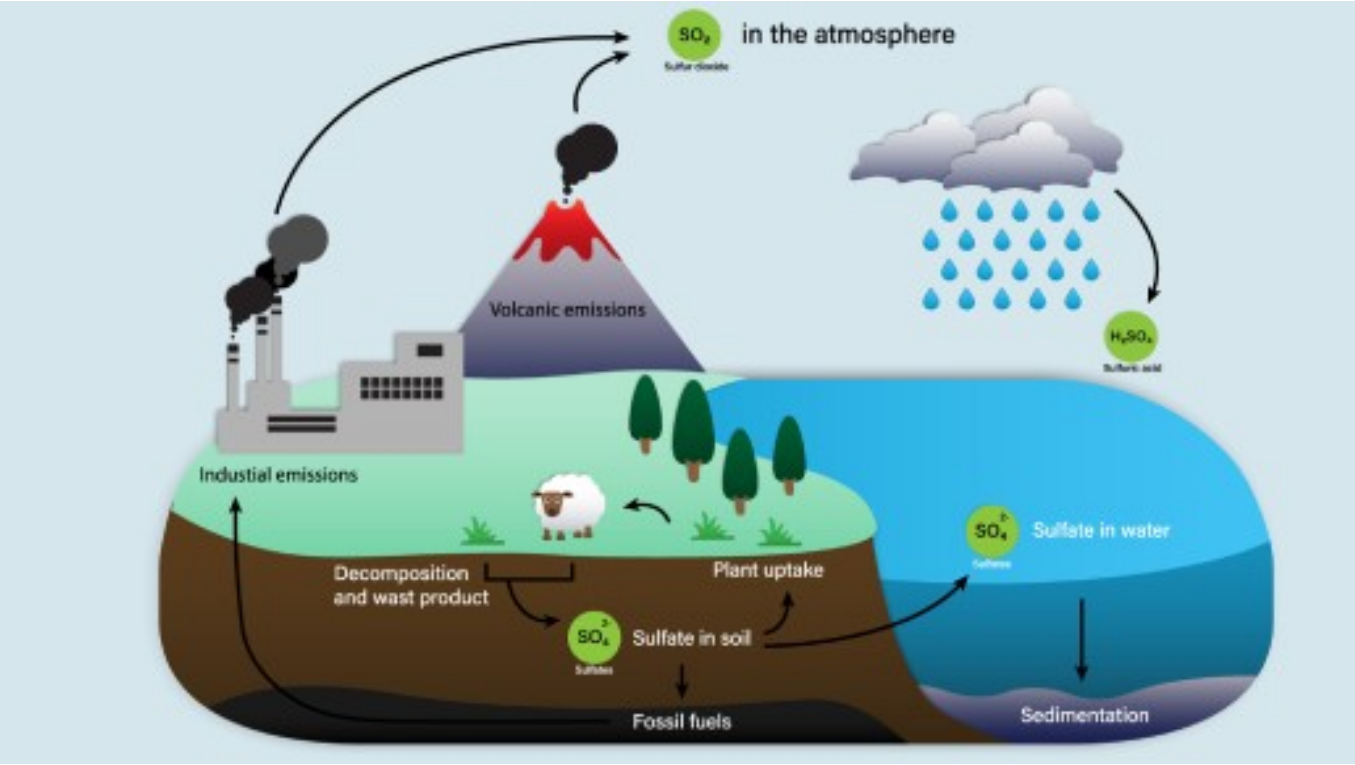
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Phosphorus Cycle

- **Weathering:** This is the breakdown of rocks and minerals at the Earth's surface through physical, chemical, and biological processes. Weathering **releases** phosphorus (and other minerals) from rocks into the soil and water.
- **Erosion:** This is the movement of weathered material (soil, rock fragments) from one place to another by natural forces like wind, water, and ice. Erosion can **transport** phosphorus-containing particles to new locations, including bodies of water.
- **Assimilation:** This is the process by which living organisms **incorporate** nutrients, like phosphorus, into their cells and tissues. Plants assimilate phosphorus from the soil through their roots, and animals assimilate it by eating plants (or other animals).
- **Decomposition:** This is the breakdown of dead organisms and organic matter by decomposers (bacteria and fungi). Decomposition **releases** phosphorus (and other nutrients) back into the soil, where it can be used by plants again.
- **Sedimentation:** This is the process by which particles of sediment (including phosphorus-containing minerals) settle out of water or air and accumulate over time, forming layers of sediment. Over long periods, these sediments can become rock, trapping the phosphorus within them until weathering releases it.

Sulfur Cycle

- The sulfur cycle involves the movement of sulfur through the atmosphere, oceans, land, and living organisms.
- Sulfur is a component of some amino acids and proteins; plays a role in enzyme activity and protein structure.
- Sulfur exists in the atmosphere as sulfur dioxide (SO_2).
- **Key Processes:** Weathering, volcanic activity, assimilation, decomposition, oxidation, reduction.



Sulfur Cycle

- **Volcanic Activity:** Volcanoes **release** large amounts of sulfur dioxide (SO_2) into the atmosphere, which can then be converted into sulfuric acid (H_2SO_4) and **return** to Earth as acid rain.
- **Assimilation:** Plants **absorb** sulfur from the soil and incorporate it into proteins and other organic molecules. Animals then **assimilate** sulfur by eating plants (or other animals).
- **Decomposition:** When organisms die, decomposers break down their remains, **releasing** sulfur back into the soil in various forms.
- **Oxidation:** This is a chemical process where **sulfur reacts with oxygen**, changing its form.
  - Example: Hydrogen sulfide (H_2S) can be oxidized to sulfur dioxide (SO_2).
- **Reduction:** This is the opposite of oxidation, where sulfur gains electrons and changes to a less oxidized form.
  - Example: Sulfates (SO_4^{2-}) can be reduced to sulfides (S^{2-}) by certain bacteria in anaerobic (oxygen-poor) environments.